

STRATEGIC PRODUCT PLACEMENT ANALYSIS (SPPA)

Project Report

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Abstract

Strategic Product Placement Analysis (SPPA) is a Business Intelligence (BI) project that examines the relationship between retail product placement and sales performance using Tableau. A retail dataset sourced from Kaggle was imported into Tableau Desktop for preprocessing, exploration, and visualization. Interactive dashboards and a Tableau Story were developed to analyze key factors including pricing, promotional effectiveness, competitor pricing, product positioning, seasonal demand, customer demographics, and overall sales performance. The final dashboards and story were published to Tableau Public and embedded into a responsive Bootstrap website built with HTML5, CSS3, and Visual Studio Code. This project illustrates how BI tools can transform raw retail data into actionable business insights.

1. Introduction

In today's data-driven retail environment, Business Intelligence plays a crucial role in enabling organizations to make informed decisions. By converting raw transactional data into meaningful information, BI tools help managers identify patterns, optimize strategies, and improve operational efficiency.

In this project, Tableau Desktop was employed to analyze a retail dataset and investigate how pricing, promotions, competitor pricing, product positioning, and seasonal demand influence sales. The project follows the complete BI workflow—from data collection and preprocessing to visualization, dashboard creation, story development, publishing, and website integration.

2. Problem Statement

Retail stores face the ongoing challenge of determining optimal shelf positions and pricing strategies to maximize sales. Product visibility, promotional offers, competitor pricing, and customer preferences all influence purchasing behavior. However, the large volume and complexity of retail data make manual analysis inefficient and error-prone. This project addresses these challenges by leveraging Tableau to build interactive dashboards and stories that empower retailers to make data-driven decisions regarding product placement and pricing.

3. Project Objectives

- Collect a retail dataset from Kaggle in CSV format.
- Import and connect the dataset to Tableau Desktop.
- Preprocess, validate, and format the data for analysis.
- Create a range of meaningful visualizations (bar charts, scatter plots, heat maps, etc.).
- Develop an interactive dashboard that consolidates all visualizations.
- Build a Tableau Story for sequential, presentation-friendly analysis.
- Publish the completed workbook to Tableau Public.
- Embed the dashboard and story into a responsive Bootstrap website using HTML5, CSS3, and Visual Studio Code.

4. Scope of the Project

This project focuses on analyzing historical retail data to identify sales trends, evaluate pricing and promotional strategies, and understand customer behavior. The interactive dashboards enable retail managers to filter data and explore insights dynamically, facilitating better product placement decisions. The methodology and workflow are scalable and can be extended to real-time business environments.

5. Technology Stack and Tools

Component	Technology Used
Dataset Source	Kaggle
Data Processing & Visualization	Tableau Desktop, Tableau Public
Web Development	Visual Studio Code, HTML5, CSS3, Bootstrap 5
Browser Testing	Google Chrome
Video Sharing	Google Drive

6. Dataset Collection and Description

6.1 Collection

The dataset was downloaded from Kaggle in CSV format. It comprises approximately 1,000 product records with attributes that support comprehensive retail analysis.

6.2 Description

The dataset includes both numerical and categorical fields, structured as follows:

- **Dimensions (Categorical):** Product Category, Promotion Status, Product Position, Customer Demographics, Seasonal Demand
- **Measures (Numerical):** Product Price, Competitor Price, Sales Volume, Foot Traffic

7. Data Preparation and Preprocessing

The CSV file was imported into Tableau Desktop. Data exploration confirmed that all 1,000 records were successfully loaded. Basic cleaning involved checking for missing values and duplicates. Calculated fields were established to provide deeper insights:

Calculated Field	Purpose
Price Difference	Compares product price with competitor price
Promotion Status	Converts raw promotion values into user-friendly labels
Discount Category	Groups products based on price differences
Sales per Price	Measures sales efficiency relative to product price

8. Visualizations Created

No.	Visualization Type	Purpose
1	Bar Chart	Sales Volume by Product Category – Top-performing categories
2	Stacked Bar Chart	Promotion Analysis – Promoted vs. non-promoted comparison
3	Donut Chart	Product Position Distribution – Shelf placement patterns
4	Scatter Plot	Competitor Price Comparison – Pricing opportunities
5	Line Chart	Seasonal Demand Analysis – Sales fluctuations over time
6	Pie Chart	Sales by Customer Demographics – Key customer segments
7	Bubble Chart	Foot Traffic vs. Sales Volume – Traffic influence on sales
8	Heat Map	Product Placement Heat Map – Performance by position/category

9. Dashboard Development

All eight visualizations were combined into a single interactive Tableau Dashboard. Key design elements included layout containers for structured arrangement and interactive filters (Product Category, Promotion Status, etc.) that dynamically update all visualizations simultaneously. This ensures that retail managers can drill down into specific data points without losing context.

10. Story Development

A Tableau Story was created to present the analysis in a narrative format. Story points guide stakeholders through pricing analysis, sales trends, customer demographics, and promotional effectiveness using captions and annotations. This approach translates complex visual data into a logical business argument.

11. Publishing & Integration

The completed workbook was published to Tableau Public. These interactive components were then embedded into a responsive project website built with HTML5, CSS3, and Bootstrap 5. This final step ensures stakeholders can access the data insights from any device, including tablets and smartphones.

12. Results and Discussion

Key Findings

- **Product Placement:** Shelf position has a significant impact on sales volume. High-visibility zones correlate strongly with higher turnover rates.
- **Promotions:** Promotional campaigns positively affect several product categories, particularly during seasonal peaks.
- **Competitor Pricing:** Price comparison analysis reveals specific price points where the store can adjust margins to gain a competitive edge.
- **Seasonal Demand:** Clear demand fluctuations underscore the importance of dynamic inventory planning.
- **Customer Demographics:** Demographic insights support more effective, targeted marketing strategies based on regional customer bases.

13. Conclusion

This project successfully demonstrates the complete Business Intelligence lifecycle. Tableau effectively transformed raw retail data into meaningful insights that support informed decision-making. By integrating these visualisations into a web platform, the analysis becomes an accessible tool for strategic planning rather than just a static report.

14. References

- Kaggle – Dataset Source
- Tableau Documentation & Learning Resources
- Bootstrap 5 Documentation
- MDN Web Docs (HTML5 and CSS3)
- Visual Studio Code Documentation